Multiple-Answer Multiple Choice Questions
2D Taylor Series Expansion

Department of Mathematics
Multivariable Calculus

Abstract

This document presents multiple-answer multiple choice questions on 2D Taylor Series Expansion. Each question may have **one or more correct answers**. Students must identify all correct options. Complete solutions and explanations are provided.

Instructions

- Each question may have **one or more correct answers**
- Mark **ALL correct answers** for each question
- There is **no partial credit** for incomplete answers
- Use the space provided for calculations
- Total: 10 questions

1 Basic Concepts and Definitions

Question 1

For the 2D Taylor expansion of $f(x,y)$ about (a,b), which statements are TRUE?

A) The linear approximation is: $L(x,y) = f(a,b) + f_x(a,b)(x-a) + f_y(a,b)(y-b)$

B) The quadratic approximation is: $Q(x,y) = L(x,y) + \frac{1}{2}f_{xx}(a,b)(x-a)^2 + f_{xy}(a,b)(x-a)(y-b) + \frac{1}{2}f_{yy}(a,b)(y-b)^2$

C) If all second partials are continuous, then $f_{xy} = f_{yx}$ and the mixed term can be written as $f_{xy}(x-a)(y-b)$

D) The remainder term $R_2(x,y)$ satisfies $\lim_{(x,y)\to(a,b)} \frac{R_2(x,y)}{\|(x-a,y-b)\|^2} = 0$

Correct Answer(s): $\boxed{A, B, C, D}$
Explanation: All statements are correct definitions and properties of 2D Taylor expansion.

Question 2

For $f(x, y) = e^{xy}$ expanded about $(0, 0)$, which terms appear in the expansion up to third order?

A) Constant term: 1

B) Linear term in x only: x

C) Quadratic term: $\frac{1}{2}x^2y^2$

D) Mixed quadratic term: xy

Correct Answer(s): $\boxed{A, D}$
Explanation: Compute derivatives:

$$\begin{aligned} f(0,0) &= 1 \\ f_x = ye^{xy} &\Rightarrow f_x(0,0) = 0 \\ f_y = xe^{xy} &\Rightarrow f_y(0,0) = 0 \\ f_{xx} = y^2e^{xy} &\Rightarrow f_{xx}(0,0) = 0 \\ f_{xy} = e^{xy} + xye^{xy} &\Rightarrow f_{xy}(0,0) = 1 \\ f_{yy} = x^2e^{xy} &\Rightarrow f_{yy}(0,0) = 0 \end{aligned}$$

So expansion: $1 + xy + \cdots$. Option C is 4th order, not quadratic.

2 Applications and Specific Functions

Question 3

Which of the following functions have the SAME second-order Taylor expansion about $(0,0)$?

A) $f(x,y) = \cos(x+y)$

B) $g(x,y) = 1 - \frac{1}{2}(x+y)^2$

C) $h(x,y) = \cos x \cos y - \sin x \sin y$

D) $k(x,y) = e^{-\frac{1}{2}(x^2+2xy+y^2)}$

Correct Answer(s): $\boxed{A, B, C, D}$
Explanation: Compute expansions:

- $\cos(x+y) = 1 - \frac{1}{2}(x+y)^2 + \cdots$
- $1 - \frac{1}{2}(x+y)^2$ is exactly quadratic part
- $\cos x \cos y - \sin x \sin y = \cos(x+y)$ (trig identity)
- $e^{-\frac{1}{2}(x^2+2xy+y^2)} = 1 - \frac{1}{2}(x^2+2xy+y^2) + \cdots$

All have same quadratic approximation.

3 Computing Expansions

Question 5

For $f(x, y) = \ln(1 + x + y)$ about $(0, 0)$, which terms appear correctly in its expansion?

A) Constant: 0

B) Linear: $x + y$

C) Quadratic: $-\frac{1}{2}(x^2 + 2xy + y^2)$

D) Cubic: $\frac{1}{3}(x^3 + 3x^2y + 3xy^2 + y^3)$

Correct Answer(s): $\boxed{A, B, C, D}$

Explanation: Let $u = x + y$. Then $\ln(1 + u) = u - \frac{u^2}{2} + \frac{u^3}{3} - \cdots$. Substitute $u = x + y$:

$$\begin{aligned}
\text{Constant} &= \ln(1) = 0 \\
\text{Linear} &= x + y \\
\text{Quadratic} &= -\frac{1}{2}(x + y)^2 = -\frac{1}{2}(x^2 + 2xy + y^2) \\
\text{Cubic} &= \frac{1}{3}(x + y)^3 = \frac{1}{3}(x^3 + 3x^2y + 3xy^2 + y^3)
\end{aligned}$$

All options are correct.

Question 6

Which properties must a function satisfy for its 2D Taylor expansion (up to order n) to be valid?

A) Function must be differentiable at (a, b)

B) All partial derivatives up to order n must exist

C) All partial derivatives up to order n must be continuous near (a, b)

D) Function must be analytic (equal to its Taylor series)

Correct Answer(s): $\boxed{A, B}$
Explanation:

- A: Needed for linear approximation
- B: Needed for nth order polynomial
- C: Needed for Lagrange remainder form, but not for existence of Taylor polynomial
- D: Analyticity is stronger than needed for finite Taylor expansion

4 Special Cases and Patterns

Question 7

For $f(x, y) = \frac{1}{1-xy}$ about $(0, 0)$, the expansion includes:

A) $1 + xy$

B) $1 + xy + x^2y^2$

C) $1 + xy + x^2y^2 + x^3y^3$

D) All terms of form $(xy)^n$ for $n = 0, 1, 2, \ldots$

Correct Answer(s): $\boxed{A, B, C, D}$
Explanation: Recognize geometric series:

$$\frac{1}{1-xy} = 1 + xy + (xy)^2 + (xy)^3 + \cdots = \sum_{n=0}^{\infty} (xy)^n$$

So all options are correct as they represent partial sums and the complete series.

Question 8

The second-order Taylor expansion of $f(x, y) = \sqrt{1 + x^2 + y^2}$ about $(0, 0)$ contains:

A) Constant term: 1

B) Linear terms: 0

C) Quadratic terms: $\frac{1}{2}(x^2 + y^2)$

D) Mixed term: xy

Correct Answer(s): $\boxed{A, B, C}$
Explanation: Compute derivatives:

$$\begin{aligned} f(0,0) &= 1 \\ f_x &= \frac{x}{\sqrt{1+x^2+y^2}} \Rightarrow f_x(0,0) = 0 \\ f_y &= \frac{y}{\sqrt{1+x^2+y^2}} \Rightarrow f_y(0,0) = 0 \\ f_{xx} &= \frac{1+y^2}{(1+x^2+y^2)^{3/2}} \Rightarrow f_{xx}(0,0) = 1 \\ f_{xy} &= -\frac{xy}{(1+x^2+y^2)^{3/2}} \Rightarrow f_{xy}(0,0) = 0 \\ f_{yy} &= \frac{1+x^2}{(1+x^2+y^2)^{3/2}} \Rightarrow f_{yy}(0,0) = 1 \end{aligned}$$

So expansion: $1 + \frac{1}{2}x^2 + \frac{1}{2}y^2 + \cdots$. No mixed term.

5 Advanced Concepts

Question 9

Which expansions about $(0,0)$ are CORRECT?

A) $\sin(x^2+y^2) = x^2+y^2-\frac{1}{6}(x^2+y^2)^3+\cdots$

B) $e^{x^2+y^2} = 1+(x^2+y^2)+\frac{1}{2}(x^2+y^2)^2+\cdots$

C) $\cos(xy) = 1-\frac{1}{2}x^2y^2+\frac{1}{24}x^4y^4-\cdots$

D) $\arctan(x+y) = (x+y)-\frac{1}{3}(x+y)^3+\cdots$

Correct Answer(s): $\boxed{A, B, C, D}$
Explanation:

- A: $\sin u = u - \frac{u^3}{6}+\cdots$ with $u = x^2+y^2$
- B: $e^u = 1+u+\frac{u^2}{2}+\cdots$ with $u = x^2+y^2$
- C: $\cos v = 1-\frac{v^2}{2}+\frac{v^4}{24}-\cdots$ with $v = xy$
- D: $\arctan w = w - \frac{w^3}{3}+\cdots$ with $w = x+y$

All are correct applications of 1D Taylor series composed with 2D functions.

Question 10

For a function symmetric in x and y [i.e., $f(x,y) = f(y,x)$], its Taylor expansion about (a,a):

A) Must have equal coefficients for x^n and y^n terms

B) Must have $f_x(a,a) = f_y(a,a)$ at symmetric points

C) Cannot have odd-powered terms in $(x-y)$

D) Mixed terms x^ky^l have same coefficient as x^ly^k

Correct Answer(s): $\boxed{A, B, D}$
Explanation:

- A: Symmetry implies equal coefficients for x^n and y^n
- B: By symmetry, $f_x(a,a) = f_y(a,a)$
- C: False: $f(x,y) = x^3+y^3$ is symmetric but expansion has odd powers
- D: True by symmetry: coefficient of x^ky^l equals coefficient of x^ly^k

Answer Key Summary

Q	Correct Answers	Remarks
1	A, B, C, D	All basic definitions correct
2	A, D	Only constant and mixed quadratic terms
3	A, B, C, D	All have same quadratic approximation
4	A, B, C	Remainder properties
5	A, B, C, D	Complete expansion of ln(1+x+y)
6	A, B	Minimal requirements for Taylor polynomial
7	A, B, C, D	Geometric series expansion
8	A, B, C	No mixed term due to symmetry
9	A, B, C, D	All are standard expansions
10	A, B, D	Properties of symmetric functions

Table 1: Summary of Correct Answers

Scoring Guide

- Each question: 10 points
- **Full credit (10 pts):** All correct answers selected, no incorrect answers
- **No credit (0 pts):** Any incorrect answer selected or not all correct answers selected
- **Total possible:** 100 points

Learning Objectives

- Understand 2D Taylor expansion formula
- Compute expansions for specific functions
- Recognize patterns in expansions
- Understand remainder term properties
- Apply symmetry considerations
- Connect 1D and 2D Taylor expansions

End of Examination